

of the 1990s, the Air Force came forward with a proposal for a new fighter that was estimated to cost five times more than the F-14.

Despite the greatly increased military spending under President Reagan, the Air Force actually was able to afford fewer planes than it bought under the Carter Administration.²⁴ One aerospace executive pointed out that if costs continued to escalate at the same rate, by "the year 2054, the entire defense budget will purchase just one tactical aircraft. This aircraft will have to be shared between Air Force and Navy, three and a half days each per week."²⁵

The cost of the arms race seems to have imposed even greater burdens on the Soviet Union and led to its collapse. Prime Minister Nikolai Ryzhkov, speaking in an interview in 1992, stated:

The main reason the negotiations broke through and agreement was achieved... was our excessively high expenditures on defense. We understood perfectly well that we had to put an end to this confrontation... [It was clear] that the country could not bear the share of state expenditures that existed at that time.²⁶

DISARMAMENT

President Nixon said, "The adversaries in the world are not in conflict because they are armed. They are armed because they are in conflict and have not yet learned peaceful ways to resolve their conflicting national interests."²⁷ The evaporation of the arms race following changes in the Soviet Union suggests that Nixon was right. But moving from stopping arming to starting disarming has not been automatic.

Indeed, what seems to be one of the simplest proposals for preventing war turns out on closer examination to be one of the most complex. Defining disarmament is not hard; it means the reduction or elimination of weapons. As a means of preventing war it is logically unassailable. Without the means to fight, you cannot have a war, any more than you can have highway accidents without vehicles. The problem comes when you try to describe precisely what weapons you want states to eliminate.

Suppose that governments say they are willing to disarm. What should we recommend? There are at least seven distinct possibilities.

1. Total disarmament. This would mean the elimination of all weapons (a "weapon" being defined as any device that could be used to kill or injure). A critic could argue that because you can strangle someone with a shoelace, shoelaces must be prohibited. Such a criticism is a frivolous version of a much more serious objection, that many devices common in modern industrial societies can be used as weapons. Fuel oil heats homes; nitrogen fertilizer increases corn yields. In combination it makes an explosive so powerful that it has replaced dynamite in many construction jobs. Commercial jetliners are widely used to transport people, messages, and goods. But if you load a Boeing 727 with fuel oil and fertilizer, you have a bomber more potent than any used in World War II. Total disarmament would mean the elimination of these and many other components of modern industrial life, in effect forcing all humanity into a primitive and frugal existence.

Even if we got rid of all weapons and all potential weapons, we could not do away with the knowledge of how to build them. Even if we get rid of

nuclear weapons, we would have to perform lobotomies on all people with advanced degrees in physics to be sure that in times of international crisis someone would not start up their production. The study of physics, engineering, and any other science would have to be strictly prohibited.

Of course we would find no support for any proposal that would lead to the eradication of the pursuit of knowledge or of modern industrial technology built on this knowledge. We would probably not even find support for the elimination of hunting rifles. And with firearms still in private possession there would be a demand for equipping law enforcement officers with firearms as well.

2. Disarmament to the lowest point consistent with domestic safety. No government leader has ever advocated disarmament that eliminated even the domestic police force. The most radical proposal seriously advanced was Point Four of Woodrow Wilson's Fourteen Points, "that national armaments will be reduced to the lowest point consistent with domestic safety." But even this proposal was too radical for states to consider. One reason is the great disparity in size among the states. A minimal police force for a large state such as Germany could threaten the security of one of its tiny neighbors such as Luxembourg and would probably make its more sizable neighbor Belgium nervous. Another reason is that states have widely varied opinions about how much domestic safety they need. England in the last century and a half has had a tradition of police armed with no more than a nightstick. France, by contrast, has had a paramilitary force, the Republican Security Companies, living in barracks, equipped with armored vehicles, and carrying firearms, including light machine guns. The French government maintains its Republic Security Companies as a riot control force, but in a totally disarmed world they would serve as an effective invading army against an England defended only by London bobbies and village constables.

3. Disarmament to the lowest point consistent with national safety. President Woodrow Wilson's Point Four was reworded at the Versailles Conference so that Article 8 of the Versailles Treaty read, "The maintenance of peace requires the reduction of national armaments to the lowest point consistent with national safety." By changing "domestic safety" to "national safety," a much wider latitude was permitted to states. Regular armies would be allowed, although only for self-defense. This goal has two problems, an old one and a new one. The old one is the question of how much force is needed for legitimate self-defense. Almost any force could be justified by that language. Do we set a basic number of 100,000 troops for each country? By those standards Luxembourg would be armed to the teeth. Do some states get larger forces because of special problems such as terrain? Poland, being flat, would need more than Switzerland with its mountains. But don't we then have to account for intentions? In recent European history the Swiss have seemed more trustworthy than the Poles.

As if these old problems weren't staggering enough, we have new ones related to modern weapons. Highly destructive nuclear weapons delivered by planes and missiles differ in fundamental ways from weapons used

in the past. They are not very useful as weapons of offense (that is, to seize and hold territory) nor are they very useful as weapons of defense (that is, to expel or hold back invaders). Most states view them as useful mainly to deter other states from using them. This view was reflected in a French statement of disarmament goals in 1991, when it finally agreed to adhere to the treaty limiting the spread of nuclear weapons. "Nuclear arsenals," the French government declared, "should be brought down to the lowest possible levels compatible with dissuasion."

But the same intercontinental missile with a nuclear warhead that dissuades another country from attacking could also be used in a disarming first strike. The bare minimum needed to deal a devastating retaliatory blow could suffice to launch a crippling first strike. How the weapons will be used depends entirely on the intentions of a state's leaders, and those intentions could change quickly, particularly if a coup or revolution brought to power a new group with aggressive intentions.

4. *Qualitative disarmament.* Because of difficulties with the general concept of national defense, proponents of disarmament have often turned to more technical discussions of types of weapons. They try to identify some weapons as offensive and eliminate them. This was attempted at the unsuccessful Geneva Disarmament Conference of 1932, which found it impossible to separate offense from defensive weapons because modern weapons can be used for both offense and defense. Recently the United States tried to make the same distinction and failed. When the U.S. occupation of South Korea ended in 1949, the United States took along with its withdrawing forces its tanks and airplanes, to keep South Korea from mounting an offensive against the North. But, as the South Koreans discovered in June 1950, without tanks and airplanes a country cannot conduct a successful defense either. Even something so obviously defensive as fortification (which cannot be moved into an enemy's country) can be part of an overall offensive plan. In 1939, Germany built a defensive line along its border with France to keep France from coming to the aid of East European allies when Germany attacked them.

One promising area for qualitative disarmament has been obnoxious weapons. The first true disarmament agreement to be reached after World War II (that is, one calling for the physical destruction of weapons) was the 1972 treaty outlawing biological weapons. By this treaty signatories agreed not only not to use such weapons in war but also to destroy existing cultures of various disease-causing organisms that could be loaded into shells and bombs. Biological weapons were easy to agree on because their military usefulness has never been demonstrated and is not believed to be great. Part of the reason is that it may take a long time for enemy troops to develop symptoms of a disease, long enough to allow them to continue to mount an offensive, for example. Another reason their use has not been attempted, at least on a large scale, is that biological agents are difficult to deliver on target. Despite science fiction fantasies, it is difficult to make people sick at a distance. The best and perhaps the only way would be a slow-flying airplane, flying over enemy troop concentrations, spraying a cloud of disease germs. Even then it is not clear how many organisms would survive the brief

trip to the ground. In any case, any military force that had the ability to fly low and slow over enemy troops would already be in a position to win the war by more conventional means.

Chemical weapons are another obnoxious weapon, but the world did not immediately follow the biological weapons treaty with a similar one to ban them. Not until 1993 were states able to agree on a Chemical Weapons Convention, which, like the Biological Weapons Treaty, prohibited even the development and production of these weapons. Even then, states were slow to ratify it. Unlike biological weapons, chemical ones have demonstrated their usefulness, most recently in the 1980s in the war between Iraq and Iran. Also, it is difficult to assure compliance with a ban. Even if stocks of chemical weapons are destroyed, new ones can easily be manufactured with equipment and plants used for similar chemicals, such as insecticides or pharmaceuticals.

The most obvious candidate for qualitative disarmament is the nuclear weapon, because more than anything else it has made international politics a serious concern for so many people. In the early 1960s the "ban the bomb" movement was active, contributing the superimposed semaphore signals for *N* and *D* (standing for nuclear disarmament), which have become the emblem of the peace movement. But during the decades of the Cold War, some of the more thoughtful peace activists turned their attention away from the issue of immediate nuclear disarmament. For example, Seymour Melman, a professor of engineering who wrote many thoughtful articles on the arms race, proposed reducing nuclear warheads to the number necessary to destroy all U.S. and Soviet cities with more than 100,000 people (a total of 150 in the United States, 175 in the Soviet Union). Melman claimed that leaders "who would not be deterred from nuclear military initiative by the prospect of destruction of these 150 cities would be too insane to be deterred by anything."²⁸ He was probably correct, but he was implicitly accepting the continued existence of nuclear weapons. Only with the demise of the Soviet Union did qualitative proposals for nuclear disarmament, such as "zero ballistic missiles," begin to make their appearance.²⁹

5. *Quantitative disarmament.* Instead of seeking to eliminate one type of weapon, as the qualitative approach recommends, states could agree to cut back numbers of one or more types of weapons, or they could agree to preserve a ratio of forces. This attention to numbers of weapons, or a quantitative approach, was used successfully in 1922 at the Washington Naval Conference. The five major powers agreed to reduce the number of battleships each had and to preserve the same ratio when they built new ships to replace old ones. The ratio was 5:5:3:1.67:1.67. The United States and Britain had the largest number; Japan was in the middle; and France and Italy had the smallest number.

The agreement, though successful, had two undesirable effects. One was to channel competition into nonregulated fields. The following years saw the states turn to building submarines and cruisers instead of battleships. The other effect was to preserve the status quo. Japan, relegated to a secondary place (only three battleships for every five the United States had),

if power grows, you can withdraw → unstable treaty

was unhappy and in 1934 demanded that the ratio be changed to allow Japan to build more battleships. When the other powers refused, Japan gave notice that it would refuse to renew the pact.

6. **Budgetary limitation.** Frustrated by the difficulties of either qualitative or quantitative limitations on weapons, states have from time to time turned to an indirect approach: limiting the amount of money states can spend on weapons. At the Hague Conference of 1899, Russia proposed a five-year freeze on all military budgets. The Germans were quick to object. Russia, they pointed out, already had superiority in numbers of soldiers. Russia could use the money saved from military budgets to build more railroads, which, although technically nonmilitary, could be used to transport Russian soldiers more quickly to the front in the event of war.³⁰

At the Geneva Disarmament Conference of 1932 the issue of budgetary limitation again came up. In preparation for the conference, a commission compiled statistics on military spending around the world. But the conference never got to make use of them. Again, it was Germany that objected, although this time for different reasons. German armaments already were limited in quantity by the Versailles Treaty; Germany hoped to make up for numerical inferiority in improved quality, which would require more expenditure.³¹

The idea of budgetary limitation has been advanced from time to time since World War II. In 1973 the Soviet foreign minister proposed that the major states of the world cut their military budgets by 10 percent and that 10 percent of these savings be donated to developing countries.³² Yet, as the Soviet Union later admitted, their published budget figure showed only a fraction of their true military spending. It included only personnel, maintenance, and construction, while omitting research, development, and procurement of new weapons.³³ Even in the relatively open society of the United States, Congress sometimes has been slow to discover that agencies in the executive branch have been violating financial limits Congress has tried to impose.³⁴ It was estimated that the Reagan Administration at one time had as much as 10 percent of military spending in "black" or secret programs.³⁵

7. **Regional disarmament.** Finding general disarmament too difficult, states have occasionally tried regional disarmament. Some agreements of this type are in effect right now. They affect Antarctica, outer space, and the seabed. These were easy cases. Very little armament was going on anyway in those areas, and they were not inhabited by people who could claim they needed arms for self-defense. There have been proposals to extend the concept of disarmament by geographic area to other regions—the Indian subcontinent, Latin America, central Europe, Africa. None of them has yet shown any success, though some Latin American states have taken a first step in this direction.

In February 1975, eight Latin American countries signed the Ayacucho Pact, agreeing to limit their acquisition of offensive weapons. But since the signing of the pact there has been no follow-up meeting. Peru, which initiated the agreement, continued to order weapons, including armored personnel carriers, tanks, and submarines. Within three years, Argentina had doubled its military budget. Neither the similarities among these countries nor their common interest in insulating their region from outside pressures

was strong enough to overcome the traditional problems in implementing a disarmament agreement?³⁶

A HISTORY OF DISARMAMENT AGREEMENTS

Disarmament proposals are relatively recent additions to the history of the search for peace. Only in recent times have weapons become so specialized that they are clearly distinguishable from the tools of everyday life. The Bible talks of beating swords into plowshares and plowshares back into swords.³⁷ In Biblical times rearmament could be accomplished overnight. The complexity of modern weapons makes this no longer possible. The disarmament movement came into being only after the Industrial Revolution had created weapons whose complexity gave rise to talk about abolishing weapons.

The late nineteenth century saw a number of organized attempts to get states to reduce their arms, culminating in the conferences at The Hague in 1899 and 1907.³⁸ The Hague Conferences did provide for laws regulating the conduct of war but did not achieve any reduction or elimination of arms. What is often described as the only successful disarmament agreement of the nineteenth century goes back much earlier. This was the Rush-Bagot Agreement of 1817 between the United States and Britain.

The Rush-Bagot Agreement provided for disarmament on the Great Lakes. Each country agreed to have only three ships, of equal size and armament. Contrary to what many schoolbooks say, the agreement was violated repeatedly. The worst violations occurred in the late 1830s, when the British sent armed ships into the Great Lakes to mount retaliatory raids against American sanctuaries for guerrilla parties who were raiding Canada. Land fortifications were maintained until 1871, when the United States signed a treaty with Britain settling claims arising from the Civil War. Only then was disarmament extended gradually and without formal agreement to the long land border. Not until the twentieth century, when the possibility of war between Canada and the United States finally disappeared from the minds of policy-planners, did violation cease.

The Rush-Bagot Agreement is sometimes cited to prove that disarmament agreement leads to peaceful relations between states. A close examination of the historical record seems to support the opposite view: Only when there are already peaceful relations among states is a disarmament agreement workable. An agreement may exist on paper, such as the Rush-Bagot Agreement did, but it is in danger of violation at any crisis.³⁹

The period after World War I saw two kinds of disarmament. One was the disarmament forced on the losing countries by the winners. This was justified as the first step toward general disarmament by all European states. But as the years passed it became clear that the victors did not intend to disarm down to the levels they had imposed on Germany. As a result, the Germans began to insist on the right to arm up to their level. In the end, disarmament only exacerbated the tensions that brought about World War II.

The other kind of disarmament was voluntary, and it met with some success, at least in the beginning. One consequence of World War I was that the United States developed its military potential. Despite the fact that the United States was an ally of

Britain's, the British looked on increased U.S. naval power as they did upon increases by any country: with grave suspicion. Partly because of this budding conflict, a naval conference was convened in Washington in 1921. It resulted in the Washington Naval Treaty of 1922, mentioned earlier, which reduced the number of battleships.

It is hard to decide what lesson to draw from the 1922 agreement. Some have argued that it proves disarmament possible.⁴⁰ Others have used it to illustrate the point that disarmament can be limited to a region or to a type of weapon but cannot be both general (applying to all countries) and complete (covering all weapons).⁴¹ The most pessimistic observers have mentioned that the battleship was already considered obsolescent by the time the conference convened, and the participants had no plans to build more ships in any case.⁴² Even by this interpretation, the Washington Naval Treaty did contribute something: At least states did not hang on to outdated weapons just because all other states did. But that contribution was minute.

Even this type of modest contribution was no longer possible by the time of the Geneva Disarmament Conference of 1932. The Geneva Conference was the long-delayed fulfillment of the promise of the Versailles settlement that German disarmament would be the first step toward general disarmament. By the time the conference convened, it was too late for that. The conference failed for both technical and political reasons. Technically, it proved impossible to agree on the questions raised by new weapons. How were tanks and airplanes to be treated? As offensive weapons or as defensive ones? Politically, the conflict between France and Germany was too severe. Germany insisted on equality of armaments with France; France insisted on security from attack by Germany. Any slim hope of compromise was ended by the accession to power of the Nazis in January 1933.

The period after World War II again saw two kinds of disarmament, forced and voluntary. Germany and Japan were forcibly disarmed in consequence of their losing the war. But events in international politics led to very different outcomes this time. The United States' perception of a Soviet threat led it to take the initiative in urging rearmament of both these countries within less than ten years. The most dramatic shift in policy was toward Japan. As part of the American occupation, the Japanese had been given a new constitution that was in effect dictated by the head of the occupation, General Douglas MacArthur. MacArthur had insisted on including Article IX, that "the Japanese people forever renounce war as a sovereign right of the nation and the threat or use of force as a means of settling international disputes." To accomplish this, the article states, "land, sea, and air forces, as well as other war potential, will never be maintained." But in 1950 all four U.S. occupation divisions in Japan were transferred to fight in Korea, and General MacArthur insisted that the Japanese form a National Police Reserve Force of 75,000 men, equipped with tanks, mortars, and machine guns. In 1954 this became the Self-Defense Force. Today, with 250,000 men, it is larger than the defense forces of most other states.⁴³

Voluntary attempts to reduce or abolish arms met with failure in the years following World War II. A number of plans were advanced, but none was even accepted as a talking point. The failure of disarmament as an approach to peace during these decades is best illustrated by the official positions of both the United States and the Soviet Union. Both claimed to favor a utopian goal of "general and complete disarmament" (meaning all kinds of weapons and all countries). In a draft treaty submitted to the United Nations in 1962, the United States specified what it meant: elimina-

tion of all weapons of mass destruction (nuclear, chemical, and biological); elimination of all nonnuclear forces except (1) those needed for internal security, and (2) agreed personnel for a UN peace force.⁴⁴

A turn toward serious negotiations on disarmament in the 1980s was signaled by the abandonment of such grandiose proposals and their replacement by more limited goals. In January of 1986, General Secretary Gorbachev proposed that within next 5 to 8 years, the Soviet Union and the United States each reduce by half the number of its nuclear weapons that could reach the other's territory. This more realistic goal proved feasible.

As a result of Gorbachev's more conciliatory approach, the United States and the Soviet Union were able to conclude in December 1987, the first agreement since the Cold War began that called for actual disarmament of nuclear weapons. The Intermediate Nuclear Forces (INF) Treaty called for the destruction not of all missiles but of those with ranges less than inter-continental range—between 500 kilometers (311 miles) and 5,500 kilometers (3,418 miles). Both sides accepted mandatory, on-site verification. Each country opened its bases and manufacturing sites to inspections from the other country and allowed observers to watch the missiles being destroyed. Each side was given the right to conduct inspections on very short notice, or, in other words, to make spot checks. Such monitoring is sometimes called "intrusive" because it allows rivals access to normally secret facilities. For years, failure to agree on on-site inspection was a major stumbling block, and its acceptance as part of the INF marked a major breakthrough.

The INF Treaty was followed by the Strategic Arms Reduction Treaty (START), signed on July 31, 1991, which called for the destruction of some of the long-range nuclear weapons possessed by each side. President Gorbachev took the occasion to declare the arms race ended, saying, "Thank God, as we say in Russian, we stopped this."⁴⁵ Under START, the Soviet Union was to drop from about 11,000 to 7,000 long-range nuclear weapons, the United States from 12,000 to 9,000.

Agreement to reduce nuclear weapons was accompanied by a Treaty on Conventional Forces in Europe in 1990. The treaty required the Soviet Union and its allies to destroy weapons down to the level of the United States and its allies. The ceiling on tanks, for example, was fixed at 20,000 for each side, compared to the original NATO number of 22,000 and the Warsaw Pact number of 41,000.

OBSTACLES TO DISARMAMENT: INSPECTION

Once a new set of leaders in the Soviet Union decided they needed to end the economic burden of the arms race, agreement came fairly rapidly. Just as important as the content of the specific disarmament treaties was the background of an entirely new Soviet orientation represented by such words as *glasnost* (openness) and *perestroika* (reform). But not all states have undergone such political shifts, and states such as Iraq, North Korea, and China are difficult partners in disarmament negotiations. Obstacles still remain.

One of these obstacles is inspection to ensure compliance with agreements. The inspection issue has been a troublesome one since people first proposed disarmament. States arm because they distrust each other. Once they begin to disarm, this distrust continues, at least until some evidence to the contrary has been accumulated.

Any state that distrusts its rivals enough to arm against them will distrust them enough to require more than their word that they are disarming.

Recent years have provided evidence that inspection is necessary. The 1972 Biological Weapons Treaty rested on trust alone. It included no provisions for inspection, in part because the U.S. government assumed the weapons would not be useful even if they were developed. We now know that the Soviet Union violated the agreement in a massive way.

The first hint came in 1979 with reports of an outbreak of anthrax in the city of Sverdlovsk following an explosion at a military research site in that city. Although U.S. intelligence agencies were not certain at the time, it was later revealed by Russian President Boris Yeltsin that the accident resulted from a Soviet biological warfare program clearly forbidden by the treaty. Then, after the collapse of the communist regime, a former Soviet scientist revealed a Soviet germ warfare program that had employed 70,000 scientists and technicians at scores of sites, studying 50 different agents including smallpox, plague, and anthrax and preparing a dozen for use in warfare.⁴⁶

Thus, inspection is an essential component of disarmament. It is also exceedingly difficult. It requires something almost impossible to provide: positive evidence of nonactivity. Positive evidence of activity is a lot easier to acquire. The part of disarmament that calls for destroying existing weapons is fairly easy. In effect, you build a big bonfire, invite your enemies in, and throw your bombers on it. The hard part is to prove that no new weapons are being built and that no existing weapons were hidden to escape the original census that indicated how many each country had. You could have a number of sources of information—tourists, military attachés, airplane flights once a month—and still find nothing. That would not prove that hidden arsenals did not exist, only that you hadn't found any. The problem is more troubling today because of the destructive power modern weapons have. It would not matter if a country hid a couple of dozen tanks under haystacks. They would not be a big threat. A couple of dozen ICBM's with megaton warheads would be an entirely different matter.

Even when inspection is part of an agreement, we can expect that noncompliance will be accompanied by deception to make discovery more difficult. When the Germans were secretly rearming after World War I, they used a number of devices to deceive the allied commission monitoring compliance. One was to build ammunition factories in Russia in return for some of the output. Another was to develop sport flying clubs and commercial airline pilot schools to train young men for the then-forbidden air force.

One of the most ingenious (and discouraging) examples of deception occurred during the Italian war against Ethiopia in 1935 and 1936. The Italians, contrary to international law, were using poison gas. Foreign journalists discovered what the Italians were doing and photographed victims of the gas attacks. The Italian military censor, instead of forbidding the dispatch of all photographs, substituted pictures showing leprosy victims. The world press printed the substituted pictures with the original captions stating that the pictures showed gas victims. Prominent medical authorities came forward to state that the people shown in the photographs were not gas victims at all but suffered from leprosy. Not only was the truth about what the Italians were doing obscured but subsequent reports were treated with skepticism.⁴⁷

A similar problem developed for the Czechs in 1938, when Hitler began making threats over the issue of the Sudeten Germans. The alarmed Czechs accused the Germans of mobilizing against them. At that time the accusation was false, and Hitler seized the opportunity to invite French and British representatives to tour the border area. They found no signs of mobilization, and as a result French and British faith in the judgment and even in the motives of the Czechs was weakened. It became harder for the Czechs to argue their case in the following months, although we now know that Hitler had his general staff prepare plans for an invasion of Czechoslovakia in 1937.⁴⁸

With all the opportunities the violator has for manipulating inspectors, it is easy to see why inspection has been a stumbling block. Inspection is especially difficult in societies with severe restrictions on speech, the press, and even the movement of citizens. In the Soviet Union under communist rule, topographic maps of the kind available at an American stationery store were treated like military documents. Even school atlases were made deliberately inaccurate, as though they otherwise might be of some value to foreign military planners.⁴⁹ Revelations after the Gulf War showed that Iraq had manipulated international inspections of its nuclear facilities. With greater access as a result of the cease-fire resolution, UN inspection teams discovered a much greater capacity to produce fuel for bombs than Iraq had earlier acknowledged. And even with legal access, they were not sure they had discovered all of its armaments.

OBSTACLES TO DISARMAMENT: ENFORCEMENT

Inspection is such a difficult issue that negotiations have often not moved beyond it. When they do, they are confronted by a second obstacle—enforcement. What if inspection reveals a violation? Most of the time, the only remedy acceptable in international relations has been to imitate the violator. In 1921 and 1922, states meeting at the Washington Naval Conference voluntarily imposed limitations on the number of battleships they would possess. In 1934 the Japanese informed the other parties to the treaty that if Japan were not allowed equality, it would withdraw from the treaty. In the absence of a world government and a world police force, no penalty could be applied to Japan short of going to war. States could respond only by building additional ships of their own.

After World War II the situation was different. Disarmament of Japan (along with Germany) was not voluntary but imposed by the victors. Disarmament was facilitated in both cases by the total discrediting of their military establishments, the total destruction of their economies, and the physical occupation of their territory by troops. The populations themselves accepted their disarmament, and when later the United States wanted the countries to make some contribution against the Soviet threat, they resisted. Even a half-century later, neither Japan nor Germany have nuclear weapons, chemical weapons, missiles, or high-technology weapons of the latest type. Both have restraints on arms exports and on the deployment of what armed forces they have; both refused to contribute troops to the allied coalition against Saddam Hussein.

After the Persian Gulf War, in 1991 Iraq too agreed to give up weapons of mass destruction, but under conditions closer to those that prevailed after World War I

than World War II. Iraq was defeated, but its territory was not occupied and its government not replaced. The international community was able to impose its will—as had happened to Germany in 1919. According to Resolution 687, passed by the Security Council on April 3, 1991 and agreed to by Iraq two days later, the UN created a Special Commission (UNSCOM) to develop a plan for inspecting all of Iraq's weapons of mass destruction, taking charge of them, and destroying them. By accepting the resolution, Iraq agreed never again to try to develop biological, chemical, or nuclear weapons.⁵⁰

But paper agreement is one thing and actual compliance another. Despite its defeat and its declared adherence to UN conditions, Iraq cheated. In the years after the cease-fire, despite obstruction and concealment by the Iraqis, UNSCOM discovered the following: a continuing capacity to build nuclear weapons including a stockpile of weapons-grade fissile material, half the prewar chemical stocks, and some undestroyed Scud missiles—perhaps as many as 100. Iraq had promised to reveal its chemical arsenal and claimed it had fewer than 11,000 shells; inspectors discovered 46,000.⁵¹ The defection from Iraq in 1995 of one of President Saddam Hussein's sons-in-law, who had been in charge of weapons development, revealed that Iraq had also had an extensive program for developing delivery systems for germ warfare.

The case of Iraq involved the disarmament of just one country, recently defeated in war and lacking significant military power. Other countries remained armed and were not panicked by revelations of Iraq's cheating. The situation would be more serious in the event of general disarmament, a violation by one country might well lead to rearmament by rivals, and however unstable an arms race might be, a rearmament race would be even more unstable. The recent technological arms race was full of uncertainties about what weapons could be built, how well they might work, and what countermeasures the enemy might have. But in a rearmament race, none of these certainties would exist. States would know how to reproduce the weapons they earlier had destroyed, and the first state to rebuild those weapons would be tempted to use them before other states could catch up.

OBSTACLES TO DISARMAMENT: SOVEREIGNTY

Another aspect of the enforcement question is the refusal of some countries to go along with a disarmament agreement. One solution would be to include them by force, going to war if necessary to disarm them. If the countries that refuse to agree are major powers, this course might be unwise. For decades both France and China refused to sign the partial test ban on nuclear weapons. No attempt was made to force them into compliance. Had French or Chinese testing programs ever threatened the security of countries that had signed the treaty, the most likely course would have been for them to break the treaty and resume testing themselves.

Even when the countries that refuse to go along are minor powers, adherence is almost never enforced. The only exceptions occur after a defeat in war. Germany was required to abandon chemical weapons after World War I; Iraq was required to abandon chemical, biological, and nuclear weapons after the Gulf War of 1991.

Otherwise, small states in the exercise of their sovereignty can refuse to go along. Despite alarm about the spread of nuclear weapons, small states such as India, Pakistan, Israel, and South Africa refused to sign the Nuclear Non-Proliferation Treaty

of 1968. Often states will argue that such agreements are unfair, because they tend to support the status quo. An Indian diplomat, noting that the Non-Proliferation Treaty did not require the major powers to give up their nuclear weapons, called it an agreement on prohibition dictated by alcoholics.⁵² India not only refused to sign it but went ahead and developed its own nuclear explosive device, which it exploded in 1974.

The problem is not a new one. Disarmament proposals in the nineteenth century ran into the same difficulty. Prussia under Bismarck, a country that wanted to change the status quo in the 1870s, resisted the appeals of various disarmament groups most vigorously.⁵³ Furthermore, even if some of the nineteenth-century proposals had been adopted, they would have permitted colonial wars in the name of maintaining "internal order."⁵⁴

The entrance of former colonial possessions into world politics has compounded the problem of arriving at disarmament agreements. Not all these countries put the avoidance of war at the top of their priorities. Saddam Hussein took Iraq into war against Iran in 1980 and against Kuwait in 1990. Cuba's Che Guevara said in 1962, "We must proceed along the path of liberation, even if that costs millions of atomic victims."⁵⁵ Horrifying as that statement might be to Americans, whose own resort to arms for liberation occurred over two centuries ago, it is a reminder that disarmament proposals freezing the status quo will not find universal acceptance.

Because an inferior side is reluctant to make its inferiority permanent, we might expect disarmament to be easiest when two sides approach equality. Unfortunately even this does not always work. The distrust that permeates international relations sometimes makes agreements impossible under the best of circumstances. In 1907 the British were leading in the *Dreadnought* competition, having demonstrated their resolve not to let the Germans or anyone else threaten their naval supremacy. Having made their point, the British offered to cut back on *Dreadnoughts* if other countries would follow. For the Germans this cutback would have been a sensible move because Germany was basically a land power, facing its principal danger from enemies on the continent. But the proponents of the German navy seized on this British initiative to argue their case more fervently. "The British economy must be hurting," they cried. "Now is the time to push harder."⁵⁶ Equality or near equality can be as much a spur to increased effort as an incentive to agree.

A greater gap separated the Americans from the rest of the world in 1945, when they alone had atomic weapons. It was thus natural that they viewed their proposal to submit this weapon to international control as a generous move. The U.S. plan, drawn up by Dean Acheson and David Lilienthal but often called the "Baruch Plan" after the man who submitted it, called for establishing an international authority with the right to inspect production facilities around the world. After this authority was in operation, the United States would destroy its atomic weapons. The Russians rejected the Baruch Plan and offered a counterproposal: The United States would destroy its atomic weapons first, and then inspection would be permitted.⁵⁷ In each case the proposal was framed in such a way that the other side would have to make the first move in giving up whatever gave it an advantage.

This pattern has been repeated often enough to lead some observers to conclude that disarmament proposals have more often than not simply been moves in the armament game.⁵⁸ The problem was often illustrated during the Cold War years

by a modern animal fable. The members of the animal kingdom gathered for a disarmament conference. The eagle asked for the abolition of fangs. The lion asked for the abolition of beaks and tusks. The elephant asked for the abolition of talons and teeth. Finally, the bear in tones of sweet reasonableness asked for the abolition of all weapons. Quarrels, he said, should be settled by hugging.⁵⁹

PROSPECTS FOR DISARMAMENT

The changed conditions in world politics changed some states' definitions of their interests. One of the major problems with arms races is accelerating costs. Nowhere is this more apparent than in the cost of aircraft. In wrestling with its military budget, the United States tried to deal with several programs for "the next generation" of fighters. The Air Force's advanced fighter aircraft program, the F-22, would cost \$95 billion for 650 planes, or \$146 million apiece.

Advanced fighter aircraft seem like a good candidate for an arms control agreement. All countries could agree not to pursue such expensive programs, thereby saving everybody a lot of money. The traditional argument of a Soviet threat no longer is convincing. But as the old argument for buying such weapons disappeared, a new one appeared. It was precisely such high-performance aircraft that allowed the United States to triumph so easily over Saddam Hussein. Without continued development in this area, eventually that advantage would fade. Continuing to arm thus seems like a better guarantee of keeping the peace than disarmament. Furthermore, American weapons are superior and highly sought after in world markets. From 1974 to 1989, the oil-producing countries of the Middle East purchased \$220 to \$250 billion worth of weapons.⁶⁰ By agreeing not to develop an advanced fighter, the United States would be denying itself a much needed commercial advantage.

Even if several countries agreed, countries would be tempted to defect. The situation would resemble Rousseau's stag hunt. The profits from cheating would be great, and if cheating occurred, eventually another ruler like Saddam Hussein might confront the United States with superior weapons. Thus the same logic that leads states to arm against each other can be used, even in time of peace, to justify continued arms expenditure. The general condition of international anarchy and the specific memory of aggression by such rulers as Hitler and Saddam Hussein work to keep disarmament an approach to peace that is easy to conceive of but hard to implement.

Notes

1. Strom Thurmond, letter to the editor, *The New York Times*, July 16, 1995, p. E14.
2. Quoted by Leonard Wainstein in "The Dreadnought Gap," *United States Naval Institute Proceedings* (September 1966), reprinted in Robert J. Art & Kenneth Waltz, *The Use of Force* (Boston: Little, Brown, 1971), p. 168.
3. See the experiment described by Morton Deutsch and Robert M. Krauss, "Studies of Interpersonal Bargaining," *The Journal of Conflict Resolution* 6, No. 1 (March 1962), pp. 52-76.
4. Lewis F. Richardson, "Could an Arms Race End without Fighting?" *Nature* 168 (September 29, 1951), pp. 567-568.
5. Colin S. Gray, "The Arms Race Phenomenon," *World Politics* 24, No. 1 (October 1971), p. 40.
6. Samuel P. Huntington, "Arms Races: Prerequisites and Results," in *Public Policy*, ed. Carl Friedrich & Seymour Harris (Cambridge, MA: Harvard University, Graduate School of Public Administration, 1958), p. 76.

7. Bruce M. Russett, *What Price Vigilance?* (New Haven: Yale University Press, 1970), pp. 8-9.
8. Albert Wohlstetter, "Racing Forward? Or Ambling Back?" *Survey* 22, Nos. 3 and 4 (Summer-Autumn 1976), pp. 163-217.
9. Huntington, p. 65 (see note 6).
10. Bernadotte E. Schmitt, *The Coming of the War, 1914*, Vol. 1 (New York: Charles Scribner's Sons, 1930), p. 54 fn.
11. Fritz Fischer, *Germany's Aims in the First World War* (New York: W. W. Norton, 1967), p. 36.
12. Barbara Tuchman, *The Guns of August* (New York: Dell, 1962), pp. 43-44.
13. Huntington, p. 48 (see note 6).
14. Arthur J. Marder, *From Dreadnought to Scapa Flow*, Vol. 1: *The Road to War, 1904-1914* (London: Oxford University Press, 1961), p. 57.
15. Malcolm W. Hoag, "On Stability in Deterrent Races," in *The Revolution in World Politics*, ed. Morton A. Kaplan (New York: John Wiley, 1962), pp. 338-410.
16. This point is emphasized in the writings of Colin Gray. See the two articles cited in note 5.
17. Michael Nicholson, *Conflict Analysis* (New York: Barnes & Noble, 1971), pp. 133-136.
18. Michael D. Wallace, "Racing Redux: The Arms Race-Escalation Debate Revisited," in *Prisoners of War?* Charles S. Gochman & Alan Ned Sabrosky, Lexington, MA: Lexington Books, 1990, p. 119.
19. Huntington, pp. 64-65 (see note 6).
20. A. F. K. Organski & Jacek Kugler, *The War Ledger* (Chicago: University of Chicago Press, 1980), p. 218.
21. Huntington, p. 76 (see note 6).
22. Huntington, p. 76 (see note 6).
23. Drew Middleton, "Rising Plane Costs Stir Air-Tactics Debate," *The New York Times*, December 1, 1974.
24. Testimony by Franklin C. Spinney to the President's Blue Ribbon Commission on Defense Management, *The New York Times*, December 13, 1985, p. Y11.
25. Quoted by James Fallows, *National Defense* (New York: Random House, 1981), p. 38.
26. Transcript of interview with N. I. Ryzhkov, Hoover Institution's Oral History Project, Summer 1992, p. 179, cited by William C. Wohlforth, "Realism and the End of the Cold War," *International Security* 19, No. 3 (Winter 1994/95), p. 113.
27. Commencement address at the Air Force Academy, quoted in *The New York Times*, June 5, 1969.
28. Cited by Theodore Shabad, "An American Asks Cut in Arms Race," *The New York Times*, October 31, 1971.
29. Alton Frye, "Zero Ballistic Missiles," *Foreign Policy*, No. 88 (Fall 1922), pp. 3-20.
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36. David Binder, "Eight Latin Nations Declare Intention to Limit Arms," *The New York Times*, December 13, 1974; Marvin Howe, "A Big Latin Arms Pact That May Signify Little," *The New York Times*, February 16, 1975; Alan Riding, "Mexicans Proposing Cuts in Latin Arms," *The New York Times*, December 27, 1978.
37. The better-known passage is Isaiah 2:4; it is reversed in Joel 3:10.
38. These conferences are described in Tate (see note 30).
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40. Philip Noel-Baker, *The Arms Race* (New York: Oceana Publications, 1958), p. 86.
41. Hedley Bull, "Disarmament and the International System" (review of Philip Noel-Baker, *The Arms Race*), *Australian Journal of Politics and History* 1 (May 1959).